MATH317 · NUMERICAL METHODS

Midterm 461 Solutions

2024 · 60 minutes · 25 marks

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01 Solution Q1: errors and Newton

a) Absolute error = | exact − approximate |

$$\text{Relative error} = \frac{|\ \text{exact} - \text{approximate}\ |}{|\ \text{exact}\ |}$$

x	Absolute error	Relative error
4	$\lvert 2.0000 - 2.1000 \rvert = 0.1$	$\frac{0.1}{2} = 0.05 = 5\ \%$
10	$\lvert 3.1623 - 3.1600 \rvert = 0.0023$	$\frac{0.0023}{3.1623} = 0.073\ \%$

Newton

$f'(x) = 3x^2 - 2$

$f(1.5) = -0.625$ $\qquad$ $f'(1.5) = 4.75$

$$x_1 = 1.5 - \frac{-0.625}{4.75} = 1.6316$$

$f(1.6316) = 0.0803$ $\qquad$ $f'(1.6316) = 5.9864$

$$x_2 = 1.6316 - \frac{0.0803}{5.9864} = 1.6182$$

02 Solution Q2: Floating-point

Positive → s = 0

1. $(1259)_{10} = (2353)_8 = (100\ 1110\ 1011)_2$

2. $(.125)_{10} = (.1)_8 = (.001)_2$

3. $(1259.125)_{10} = (10011101011.001)_2 = (1.0011101011001)_2 \times 2^{10}$

4. c = 10 + 127 = 137 = $(211)_8$ = $(10001001)_2$

5. $[0 \mid 10001001 \mid 00111010110010000000000]_2$

   $[0100\ 0100\ 1001\ 1101\ 0110\ 0100\ 0000\ 0000]_2$

   $= [449D6400]_{16}$

The number is positive, so the first hex digit is below 8.

03 Solution Q3: Bisection

a) u = f(0) = 1 v = f(1) = e − 4 = −1.2817

u · v < 0 → f has a root in [0, 1]

f″(x) = e^x − 8 < 0 on [0, 1] → f is concave → it crosses the axis only once

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b)  n = 0:  c = 0.5        w = f(0.5) = 0.6487

            u · w > 0   →   a = c   →   [0.5, 1]

    n = 1:  c = 0.75       w = f(0.75) = −0.1330

            u · w < 0   →   b = c   →   [0.5, 0.75]

    n = 2:  c = 0.625      w = f(0.625) = 0.3057

            u · w > 0   →   a = c   →   [0.625, 0.75]
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n	c_n	$f(c_n)$	Error
0	0.5	0.6487	0.5
1	0.75	−0.1330	0.25
2	0.625	0.3057	0.125